

Question:

A 61-year-old woman is brought to an urgent care clinic by her husband with an altered mental status. The patient’s husband says that her symptoms onset acutely 24 hours ago. He says she suddenly started to feel excessively drowsy and quickly became altered. Past medical history is significant for renal transplant 18 months ago, well-managed with immunosuppressive medication. The vital signs include: temperature 39.4°C (103.0°F), blood pressure 85/50 mm Hg, pulse 135/min and respirations 24/min. On physical examination, the patient is arousable but non-responsive to commands. Oriented x 0. Glasgow Coma Scale (GCS) score is 10/15. Multiple black skin lesions are present on the trunk. Lesions are approximately 4 cm is diameter and round with a necrotic center forming an eschar.

Laboratory findings are significant for the following:

Hemoglobin 14.2 g/dL

WBC count 3,700/μL

Neutrophils 22%

Lymphocytes 52%

Monocytes 17%

Eosinophils 5%

Basophils 4%

Platelets 179,000/μL

BUN 15 mg/dL

Creatinine 0.8 mg/dL

Blood cultures are positive for Pseudomonas aeruginosa. Which of the following factors is most likely responsible for this patient’s condition?

Answer Choices:

- A. Failure of immunosuppressive therapy
- B. Decreased interleukin-2 levels
- C. DiGeorge Syndrome
- D. Decreased phagocytic cell count

Reasoning + Answer:

<think> Reasoning process here: The patient's symptoms and laboratory findings , including the presence of black skin lesions with necrotic centers , fever , and positive blood cultures for Pseudomonas aeruginosa , suggest a severe bacterial infection . The patient's immunosuppressed status due to a renal transplant increases the risk of opportunistic infections . The patient's laboratory findings also indicate a low neutrophil count , which is a type of phagocytic cell . This low neutrophil count , combined with the presence of a severe bacterial infection , suggests a problem with the patient's ability to fight off the infection . </think><answer> D . Decreased phagocytic cell count </answer>

